



Urban Informatics

Assignment - Week 9

TYPE OF QUESTION: MCQ/MSQ

Number of questions: 12

Total marks: (4 X 1) + (8 X 2) = 20 marks

MCQ/MSQ Question

QUESTION 1: [1 mark]

In the Association Rule Learning workflow, which step involves converting the dataset into a binary (0–1) incidence matrix?

- a. Data collection
- b. Rule generation
- c. Data preprocessing
- d. Interpretation and deployment

Correct Answer: c. Data preprocessing

Detailed explanation: During data preprocessing, the transaction data is transformed into a binary matrix (0–1) representing the presence or absence of items before frequent itemset generation algorithms. (Refer to Lecture 44, Slide 4)

QUESTION 2: [2 marks]

In clustering analysis, similarity between observations is usually determined using _____, while datasets containing both numeric and categorical variables can be clustered using the _____ algorithm.

- a. Distance calculations, K-Prototypes
- b. Probability estimation, K-Means
- c. Gradient descent, Principal Component Analysis
- d. Regression analysis, K-Nearest Neighbours

Correct Answer: a. Distance calculations, K-Prototypes

Detailed explanation: Most clustering algorithms determine similarity using distance measures (such as Euclidean distance). When a dataset contains both numeric and categorical variables, the K-Prototypes algorithm is commonly used because it can handle mixed data types. (Refer to Lecture 41, Slide 3 & 4)



QUESTION 3: [2 marks]

Considering the metrics for Association rule learnings, match the metrics with their correct interpretation.

Metric	Interpretation
1. Support	A. Measures the strength of a rule.
2. Confidence	B. Frequency of appearance of an item in the dataset.
3. Lift	C. Measures how often a rule appears to be true in a dataset.

- a. 1–C, 2–A, 3–B
- b. 1–A, 2–B, 3–C
- c. 1–B, 2–A, 3–C
- d. 1–B, 2–C, 3–A

Correct Answer: d. 1–B, 2–C, 3–A

Detailed explanation: Support identifies the frequency of appearance of an item in the dataset. Confidence measures how often a rule appears to be true in a dataset. Lift measures the strength of a rule. (Refer to Lecture 44, Slide 5)

QUESTION 4: [1 mark]

Which of the following best describes Association Rule Learning?

- a. A supervised learning technique used for classification of labelled datasets
- b. An unsupervised data mining method used to identify patterns and frequent co-occurring relationships among variables in large datasets
- c. A clustering algorithm that groups data based on Euclidean distance
- d. A regression technique used to predict continuous values

Correct Answer: b. An unsupervised data mining method used to identify patterns and frequent co-occurring relationships among variables in large datasets

Detailed explanation: Association Rule Learning is an unsupervised data mining method that discovers associations and frequent co-occurring relationships among variables without labelled outputs. (Refer to Lecture 44, Slide 3).

QUESTION 5: [2 marks]

In an urban clustering study, two clusters of city zones are being merged using Ward's linkage method. Cluster 1 contains 8 urban zones and Cluster 2 contains 12 urban zones. The Euclidean distance between the centroids of the clusters is 5. Calculate the increase in within-cluster variance ΔE when these two clusters are merged.

- a. 80
- b. 60
- c. 40
- d. 120

Correct Answer: d. 120



Detailed explanation: (Refer to Lecture 43, Slide 5)

$$\begin{aligned}\Delta E &= \frac{n_1 n_2}{n_1 + n_2} \times (\text{distance})^2 \\ &= \frac{8 \times 12}{8 + 12} \times 5^2 \\ &= \frac{96}{20} \times 25 \\ &= 4.8 \times 25 = 120\end{aligned}$$

QUESTION 6: [2 marks]

A geospatial analyst is clustering urban neighborhoods based on features such as population density, land surface temperature, vegetation cover, and building height using Agglomerative Hierarchical Clustering. During experimentation, the analyst observes the following behaviors:

- Method A: Clusters sometimes form long chain-like structures, where distant neighbourhoods become part of the same cluster through intermediate ones.
- Method B: Clusters are very compact, because the distance between clusters is determined by the farthest pair of points.
- Method C: Cluster merging prioritizes minimizing the increase in within-cluster variance (WCSS).

Which of the following correctly identifies the linkage methods used?

- a. A – Single Linkage, B – Complete Linkage, C – Ward's Linkage
- b. A – Average Linkage, B – Single Linkage, C – Ward's Linkage
- c. A – Complete Linkage, B – Average Linkage, C – Single Linkage
- d. A – Single Linkage, B – Ward's Linkage, C – Average Linkage

Correct Answer: a. A – Single Linkage, B – Complete Linkage, C – Ward's Linkage

Detailed Solution: Single Linkage (Nearest Neighbor) uses the distance between the closest points and can create chaining effects. Complete Linkage (Farthest Neighbor) uses the distance between farthest points which produces compact clusters. Ward's Linkage merges clusters that cause the smallest increase in within-cluster variance (WCSS). (Refer to Lecture 43, Slide 5)

QUESTION 7: [1 mark]

Which of the following correctly describes a core point in DBSCAN?

- a. A point that lies within ϵ distance of a core point but has fewer than MinPts neighbours
- b. A point whose ϵ -neighbourhood contains at least MinPts points
- c. A point that is not reachable from any other point within ϵ distance
- d. A point that connects two clusters together

Correct Answer: b. A point whose ϵ -neighbourhood contains at least MinPts points

Detailed Solution: A core point in DBSCAN is a point whose ϵ -neighbourhood contains at least MinPts points, meaning it lies in a dense region of the dataset. (Refer to Lecture 42, Slide 7)



QUESTION 8: [2 marks]

Consider the following statements about Gaussian Mixture Models (GMM) and the Bayesian Information Criterion (BIC) used for model selection.

Statement A - BIC balances good model fit and penalizes excessive model complexity, thereby preventing overfitting.

Statement B - In GMM model selection, the model with the highest BIC value is considered the optimal model.

Statement C - Increasing the number of clusters in a GMM generally increases the number of parameters, which increases the complexity penalty in BIC.

Statement D - In GMM, the “spherical” covariance type allows each cluster to have its own full covariance.

Considering the statements given above, choose the correct option.

- a. Statements A and C are true; Statements B and D are false
- b. Statements A and B are true; Statements C and D are false
- c. Statements B and D are true; Statements A and C are false
- d. Statements A, C and D are true; Statement B is false

Correct Answer: a. Statements A and C are true; Statements B and D are false tree

Detailed Solution: BIC balances goodness-of-fit and complexity. The lowest BIC indicates the best model. More clusters mean more parameters and larger penalty term. Spherical covariance assumes equal variance in all directions (circular clusters) while full covariance allows different orientations and shapes. (*Refer to Lecture 42, Slide 4*)

QUESTION 9: [2 marks]

An urban researcher collects five variables from multiple city zones (impervious surface percentage, population density, land surface temperature (LST), albedo, air pollution index). Exploratory analysis shows that impervious surface, density, and LST are strongly positively correlated, while albedo is negatively correlated with temperature and pollution. The researcher wants to create one composite urban heat–impact indicator while avoiding multicollinearity in further regression analysis.

Which of the following statements best explains how Principal Component Analysis (PCA) can be used in the creation of the indicator?

- a. PCA selects the single variable with the highest variance and removes all others to eliminate correlation.
- b. PCA constructs orthogonal components as linear combinations of the original variables, where the first component captures the maximum variance across the correlated variables.
- c. PCA converts correlated variables into independent categorical groups based on their correlation strength.
- d. PCA ranks variables based on their correlation with temperature and removes those with lower correlations.

Correct Answer: b. PCA constructs orthogonal components as linear combinations of the original variables, where the first component captures the maximum variance across the correlated variables.

Detailed Solution: PCA transforms correlated variables into orthogonal (uncorrelated) principal



components, each being a linear combination of the original variables. The first principal component captures the maximum variance, making it suitable as a composite indicator while also reducing multicollinearity. (Refer to Lecture 45, Slide 5)

QUESTION 10: [1 mark]

Which of the following statements about the Mean Shift algorithm is correct?

- a. It requires the number of clusters to be specified before running the algorithm.
- b. It groups points by minimizing within-cluster variance similar to Ward's method.
- c. It identifies clusters by iteratively moving data points toward regions of higher data density.
- d. It forms clusters based only on the farthest distance between data points.

Correct Answer: c. It identifies clusters by iteratively moving data points toward regions of higher data density.

Detailed Solution: The Mean Shift algorithm iteratively shifts each data point toward the nearest high-density region (density peak) using a kernel density estimation (KDE) approach. It does not require specifying the number of clusters beforehand and can detect clusters of arbitrary shapes. (Refer to Lecture 42, Slide 5)

QUESTION 11: [2 marks]

Match the concepts in Column A with the correct descriptions in Column B.

Column A	Column B
A. Unsupervised Learning	1. Data point belongs to multiple clusters with varying membership
B. Hard Clustering	2. An approach where there is no predefined target variable
C. Soft Clustering	3. Each observation belongs strictly to a single group
D. Dimensionality Reduction Models	4. Technique used to uncover underlying structures in high-dimensional data

- a. A-2, B-3, C-1, D-4
- b. A-1, B-2, C-3, D-4
- c. A-2, B-1, C-3, D-4
- d. A-4, B-3, C-1, D-2

Correct Answer: a. A-2, B-3, C-1, D-4

Detailed Solution: Unsupervised Learning works with no labelled output or target variable. In hard clustering, each data point belongs to exactly one cluster. In soft clustering, data points may belong to multiple clusters with different probabilities or memberships. Dimensionality reduction models discover underlying structures and reduce complexity in high-dimensional datasets. (Refer to Lecture 41, Slide 3 & 4)



QUESTION 12: [2 marks]

A city planning department is clustering 150 urban neighborhoods using K-means based on several variables (average parking occupancy, pedestrian activity density, distance to nearest metro station, traffic congestion index). The planners compute the following observations:

- I) WCSS decreases sharply until $k = 3$ and then decreases slowly afterwards.
- II) Average Silhouette Score for $k = 3$ is 0.62.
- III) Average Silhouette Score for $k = 5$ is 0.41.
- IV) Some neighborhoods at $k = 3$ have silhouette values close to 0.

Which of the following interpretations is **CORRECT** based on the observations given above?

- a. Five clusters should be chosen because WCSS always decreases as the number of clusters increases.
- b. Three clusters are inappropriate because silhouette values close to zero indicate that K-means has completely failed.
- c. Five clusters are better because silhouette score measures only cluster compactness, not separation.
- d. Three clusters are likely appropriate because the elbow occurs at $k = 3$, and the higher silhouette score indicates better separation, although some neighbourhoods may lie near cluster boundaries.

Correct Answer: d. Three clusters are likely appropriate because the elbow occurs at $k = 3$, and the higher silhouette score indicates better separation, although some neighbourhoods may lie near cluster boundaries.

Detailed Solution: WCSS measures cluster compactness (distance of points from centroids) and it always decreases as k increases. The Elbow Method identifies where the rate of decrease slows significantly, and since the elbow occurs at $k = 3$, it suggests 3 clusters capture the main structure of the dataset. For silhouette score, $k = 3$, the value of 0.62 indicates fairly strong clustering structure, whereas, for $k = 5$, the value of 0.41, indicates weaker cluster separation. Thus, $k = 3$ produces better-defined clusters. *(Refer to Lecture 41, Slide 7 & 8)*

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